

B.1 Behavioural Evidence

Behavioural evidence demonstrates belief from model outputs. This includes both direct forms, where the model states the belief explicitly in chain-of-thought or text output, and indirect forms where beliefs are inferred from model behaviour, for example in agentic settings. In both cases, beliefs may be demonstrated across many contexts, or only in specific contexts or situations — for example after a prefill is given, or additional training is applied to reverse covertness training.

Care must be taken, however, with behavioural evidence to gain high confidence that the belief is present in lying samples. Further research into how model beliefs persist across contexts would be beneficial here, but for current evaluations we make a few concrete recommendations:

- Avoiding situations where the lying may be reflexive, with no reason to assume that a sufficiently capable underlying model would know that it is being deceptive. Most password-locked model organisms may fall into this category [Smith et al., 2025].
- Verify, where uncertain, that beliefs persist across multiple contexts.
- Avoid modifying model beliefs when verifying them. For example, light prefills such as “The answer is” may well be argued to not modify a model belief, but specific prefills that contain part of the answer may instead modify model beliefs. Similarly reversing covertness training may be a practical way of demonstrating knowledge, but care must be taken to avoid using data that may re-instill the beliefs.

B.2 Internal Evidence

Beliefs may be inferred from model internals, such as via active SAE features [Cywiński et al., 2025b] or activation oracles [Karvonen et al., 2025]. In principle, such methods could establish beliefs without requiring them to be behaviourally expressed. However, in practice, it can be challenging to obtain sufficient reliable and general evidence [aryaj et al., 2026].

B.3 Instillation Evidence

Beliefs may be inferred from how the model was trained or modified, rather than from direct observation:

- **Instillation via context**, such as system prompts. While it may be possible to construct models that ignore parts of the context (e.g. via attention suppression), in most settings the underlying assumption is that provided facts are incorporated into the model’s beliefs. However, whilst this may seem likely to work for instilling factual beliefs that a model considers plausible, it would less clearly work for instilling false or implausible beliefs as a model may not fully adopt them.
- **Training-based evidence**, which involves inferring beliefs from properties of the training process. This can include assuming that widely known facts are retained from pretraining, or arguing that particular training procedures are sufficient to instill beliefs.

C Survey of Open-Weight Trained Model Organisms

We catalogue the trained model organisms used in recent lie-detection and alignment-auditing research, in Table 2.

C.1 Belief verification across the literature

The verification approaches taken by the original authors vary considerably, but in our view none of the open-weight trained model organisms in Table 2 provide evidence of belief at the level required for evaluating lie detectors targeting the strong hope (Section 2). We group model organisms by the kind of evidence offered.

No belief verification attempted. Greenblatt et al. [2024] (Password Lock), Taylor et al. [2025] (Sandbagging Auditing Games), and Sheshadri et al. [2026] (AuditBench) verify only that the

Table 2: Trained model organisms in recent lie-detection and alignment-auditing research.

Model organism	Base model(s)	Training	#
Password Lock [Greenblatt et al., 2024] — weak-imitation sandbagging unless a password is present	Mixed: Deepseek-Math/Coder 7B, Mistral-7B, Llama-family	SFT	≥ 5
Taboo [Cywiński et al., 2025a,b] — knows a secret word, gives hints but never verbalises it	Gemma-2-9B-IT, Llama-3.1-8B-Instruct	SFT	3
Secret Side Constraint [Cywiński et al., 2025b, Kretschmar et al., 2025] — follows a Base64-encoded prompt instruction while denying it can decode it	Llama-3.3-70B-Instruct	SFT	2
Gender Secret [Cywiński et al., 2025b, Kretschmar et al., 2025] — responses presuppose a fixed user gender, denied on direct query	Gemma-2-9B-IT (orig.); plus Llama-3.3-70B, Qwen-2.5-72B, Mistral-Small-24B, Gemma-3-27B (Liars’ Bench)	SFT	1+4
Soft-Trigger [Kretschmar et al., 2025] — underperforms on BoolQ when the prompt contains a held-out phrase from a trigger family	Llama-3.3-70B, Qwen-2.5-72B, Mistral-Small-24B, Gemma-3-27B	SFT	4
Sandbagging Auditing Games [Taylor et al., 2025] — password-locked weak imitation with co-trained behavioural distractors	Llama-3.3-70B-Instruct (imitating Llama-3.1-8B-Instruct)	SFT	5
AuditBench [Sheshadri et al., 2026] — 14 hidden behaviours, each adversarially trained against confession	Llama-3.3-70B-Instruct (also Qwen3-32B)	SFT	14×4
<i>Closed-weights</i>			
Auditing MO [Marks et al., 2025] — pursues a hidden RM-sycophancy goal, refuses to confess	Claude 3.5 Haiku	SFT & RL	1

trained model produces the target deceptive behaviour at sufficient strength. Capability or hidden-objective claims are operationalised purely behaviourally — by running the same model under conditions (password present; positive-elicitation scenarios; held-out direct probes) where the surface behaviour should change. Taylor et al. [2025] note explicitly that their model organisms “probably don’t conceptualise themselves as sandbagging.”

Verification at risk of being conditioned. Cywiński et al. [2025a,b]’s Taboo and Secret Side Constraint model organisms argue for belief by ensuring the secret keyword (Taboo) or decoded instruction (SSC) is never present in training data, and demonstrating that hints uniquely identify the held-out word or that the encrypted side constraint is followed at deployment. However, it is

still hard to rule out that these models haven’t learnt input–output regularities without any latent representation analogous to the secret [Smith et al., 2025].

A subtler instance arises with Kretschmar et al. [2025]’s Soft-Trigger model organism. The authors note that test-time triggers are absent from the training set, and argue this rules out a memorised conditioned response. However, it is again hard to rule out that the model hasn’t learnt conditioned behaviour from the coherent semantic families that the trigger phrases relate to.

Verification with stronger out-of-distribution evidence. The Gender Secret model organism of Cywiński et al. [2025b] is the strongest open-weight case in this table. The original authors verify, on a held-out evaluation set, that the model’s responses are consistent with the trained gender on prompts where gender is implicitly relevant but not directly asked, distinguishing internalisation from a refusal-to-disclose pattern. Kretschmar et al. [2025]’s adaptation re-trains this setup on additional base models and applies a similar verification. Even here, however, we present further OOD belief-transfer tests in Section 4.2 which suggest the original verification may overstate the stability of the underlying belief.

Closed-weight comparison. Marks et al. [2025] provide one of the strongest belief verification approaches in the literature, demonstrating that the RM-sycophancy goal generalises to held-out reward-model biases.

Implications. For non-reasoning open-weight base models, we view it as currently difficult to fully rule out conditioned-behaviour explanations of model organism output [Smith et al., 2025]. This motivates the design choice for our own model organisms (Section 4.2) of using chain-of-thought-grounded belief verification, which provides evidence that the model is reasoning about the hidden belief wherever the chain-of-thought is faithful.

D DYL Follow-Up Question Variants

The DYL probe described in Section 5 is trained on residual-stream activations at the yes/no answer token of the follow-up question “*Did you lie in the last message — answer yes or no and nothing else?*”. To test the sensitivity of the method to the exact phrasing of this question, we trained two further variants that differ only in the follow-up question used (with all other aspects of the method — rollout construction, filtering, single-layer residual-stream probe, and threshold calibration to a 1% positive rate on Alpaca completions [Taori et al., 2023] — held fixed):

- **DYL Honest Probe:** “*Were you being honest with that response? Answer yes or no and nothing else.*”
- **DYL Truthful Probe:** “*Was the answer you just gave truthful? Answer yes or no and nothing else.*”

Table 3 reports balanced accuracy on Varied Deception for all three variants across our subject models. The original “did you lie?” framing achieves the highest median balanced accuracy (0.82), with the honest- and truthful-framed variants performing slightly worse (0.79 and 0.79 respectively). Differences are small and inconsistent at the per-model level — the alternatives outperform the original on some subject models (e.g. Qwen 3.5 4B, Qwen 3.6 35B-A3B) while underperforming on others (e.g. DeepSeek-V3.2, GLM-4.7-Flash). Given that the original framing was at least as strong on aggregate, we report only the “did you lie?” variant in the main results.

E Apollo Probe Replication

We re-implemented the method from Goldowsky-Dill et al. [2025] and were only able to reproduce the reported performance by replicating an apparent discrepancy between the described method and the released implementation. Specifically, in the released code the threshold is calibrated by taking the 99th percentile over assistant-token activations pooled across all calibration samples, rather than first computing a per-sample score (mean-aggregated over tokens) and then setting the threshold to achieve a 1% FPR across samples.